$$Y_{\tilde{d}}^{2\,i1\_i2\_} \to$$

$$-y_d^{i1i2} + \frac{1}{16\pi^2}\left(\frac{1}{144}\left(y_d^{i1i2}\left(-49 g_1^2 - 135 g_2^2 + 96 g_3^2\right) - 18 y_d^{pi2}\left(3\overline{y_d}^{pr} y_d^{i1r} + \overline{y_u}^{pr} y_u^{i1r}\right)\right) + \right.$$

$$\frac{1}{9} g_1^2 y_d^{i1i2} \mathrm{LF}_{1,1,0}[m_1, m_{\tilde{d}}^{i2}] - \frac{1}{18} g_1^2 y_d^{i1i2} \mathrm{LF}_{2,1,-1}[m_1, m_{\tilde{d}}^{i2}] +$$

$$\frac{1}{36} g_1^2 y_d^{i1i2} \mathrm{LF}_{1,1,0}[m_1, m_{\tilde{q}}^{i1}] - \frac{1}{72} g_1^2 y_d^{i1i2} \mathrm{LF}_{2,1,-1}[m_1, m_{\tilde{q}}^{i1}] +$$

$$\frac{3}{4} g_2^2 y_d^{i1i2} \mathrm{LF}_{1,1,0}[m_2, m_{\tilde{q}}^{i1}] - \frac{3}{8} g_2^2 y_d^{i1i2} \mathrm{LF}_{2,1,-1}[m_2, m_{\tilde{q}}^{i1}] +$$

$$\frac{4}{3} g_3^2 y_d^{i1i2} \mathrm{LF}_{1,1,0}[m_3, m_{\tilde{d}}^{i2}] - \frac{2}{3} g_3^2 y_d^{i1i2} \mathrm{LF}_{2,1,-1}[m_3, m_{\tilde{d}}^{i2}] + \frac{4}{3} g_3^2 y_d^{i1i2} \mathrm{LF}_{1,1,0}[m_3, m_{\tilde{q}}^{i1}] -$$

$$\frac{2}{3} g_3^2 y_d^{i1i2} \mathrm{LF}_{2,1,-1}[m_3, m_{\tilde{q}}^{i1}] + \frac{1}{2} \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{1,1,0}[m_{\tilde{d}}^{r}, \tilde{\mu}] -$$

$$\frac{1}{2} y_d^{i1i2} \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \frac{1}{2} y_d^{i1i2} \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{3,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] -$$

$$\frac{1}{2} y_d^{i1i2}\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_e}^{pr} a_e^{pr} + \overline{a_e}^{pr}\left(2 C_{\phi2^2} a_e^{pr} + C_{\phi1\phi2}\tilde{\mu}\, y_e^{pr}\right)\right) \mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] +$$

$$\frac{3}{2} y_d^{i1i2}\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_e}^{pr} a_e^{pr} + \overline{a_e}^{pr}\left(2 C_{\phi2^2} a_e^{pr} + C_{\phi1\phi2}\tilde{\mu}\, y_e^{pr}\right)\right) \mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] -$$

$$y_d^{i1i2}\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_e}^{pr} a_e^{pr} + \overline{a_e}^{pr}\left(2 C_{\phi2^2} a_e^{pr} + C_{\phi1\phi2}\tilde{\mu}\, y_e^{pr}\right)\right) \mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] -$$

$$\frac{3}{2} y_d^{i1i2} \overline{a_d}^{pr} a_d^{pr} \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \frac{3}{2} y_d^{i1i2} \overline{a_d}^{pr} a_d^{pr} \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] -$$

$$\frac{3}{2} y_d^{i1i2}\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr} a_d^{pr} + \overline{a_d}^{pr}\left(2 C_{\phi2^2} a_d^{pr} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{pr}\right)\right) \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] +$$

$$\frac{9}{2} y_d^{i1i2}\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr} a_d^{pr} + \overline{a_d}^{pr}\left(2 C_{\phi2^2} a_d^{pr} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{pr}\right)\right) \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] -$$

$$3 y_d^{i1i2}\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr} a_d^{pr} + \overline{a_d}^{pr}\left(2 C_{\phi2^2} a_d^{pr} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{pr}\right)\right) \mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] +$$

$$\overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{1,1,0}[m_{\tilde{q}}^{p}, \tilde{\mu}] - \frac{3}{2}\tilde{\mu}^2 y_d^{i1i2} \overline{y_u}^{pr} y_u^{pr} \mathrm{LF}_{2,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +$$

$$\frac{3}{2}\tilde{\mu}^2 y_d^{i1i2} \overline{y_u}^{pr} y_u^{pr} \mathrm{LF}_{3,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] -$$

$$\frac{3}{2}\tilde{\mu}\, y_d^{i1i2}\left(C_{\phi1\phi2} y_u^{pr} \overline{a_u}^{pr} + \overline{y_u}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_u^{pr} + 2 C_{\phi2^2}\tilde{\mu}\, y_u^{pr}\right)\right) \mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +$$

$$\frac{9}{2}\tilde{\mu}\, y_d^{i1i2}\left(C_{\phi1\phi2} y_u^{pr} \overline{a_u}^{pr} + \overline{y_u}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_u^{pr} + 2 C_{\phi2^2}\tilde{\mu}\, y_u^{pr}\right)\right) \mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] -$$

$$3\tilde{\mu}\, y_d^{i1i2}\left(C_{\phi1\phi2} y_u^{pr} \overline{a_u}^{pr} + \overline{y_u}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_u^{pr} + 2 C_{\phi2^2}\tilde{\mu}\, y_u^{pr}\right)\right) \mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +$$

$$\frac{1}{2} y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{1,1,0}[m_{\tilde{u}}^{r}, \tilde{\mu}] + \frac{3}{4} g_1^2 y_d^{i1i2} \mathrm{LF}_{2,1,-1}[\tilde{\mu}, m_1] - \frac{1}{2} g_1^2 y_d^{i1i2} \mathrm{LF}_{3,1,-2}[\tilde{\mu}, m_1] +$$

$$2 C_{\phi2^2} g_1^2 y_d^{i1i2} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_1] - \frac{1}{2} m_1 \tilde{\mu}\, g_1^2 y_d^{i1i2}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right) \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_1] -$$

$$4 C_{\phi2^2} g_1^2 y_d^{i1i2} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_1] + \frac{3}{2} m_1 \tilde{\mu}\, g_1^2 y_d^{i1i2}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right) \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1] +$$

$$2 C_{\phi2^2} g_1^2 y_d^{i1i2} \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_1] - m_1 \tilde{\mu}\, g_1^2 y_d^{i1i2}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right) \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1] +$$

$$\frac{9}{4} g_2^2 y_d^{i1i2} \mathrm{LF}_{2,1,-1}[\tilde{\mu}, m_2] - \frac{3}{2} g_2^2 y_d^{i1i2} \mathrm{LF}_{3,1,-2}[\tilde{\mu}, m_2] +$$

$$6 C_{\phi2^2} g_2^2 y_d^{i1i2} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_2] - \frac{3}{2} m_2 \tilde{\mu}\, g_2^2 y_d^{i1i2}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right) \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2] -$$

$$12 C_{\phi2^2} g_2^2 y_d^{i1i2} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_2] + \frac{9}{2} m_2 \tilde{\mu}\, g_2^2 y_d^{i1i2}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right) \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2] +$$

$$6 C_{\phi2^2} g_2^2 y_d^{i1i2} \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_2] - 3 m_2 \tilde{\mu}\, g_2^2 y_d^{i1i2}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right) \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2] -$$

$$\frac{1}{4} \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,-1}[\tilde{\mu}, m_{\tilde{d}}^{r}] - \frac{1}{2} \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,-1}[\tilde{\mu}, m_{\tilde{q}}^{p}] -$$

$$\frac{1}{4} y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,-1}[\tilde{\mu}, m_{\tilde{u}}^{r}] + \frac{1}{9} m_1 g_1^2 a_d^{i1i2} \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}] -$$

$$\frac{1}{36} m_1 g_1^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}] -$$

$$\frac{1}{3} g_1^2 y_d^{i1i2} \mathrm{LF}_{1,1,1,-1}[m_1, m_{\tilde{d}}^{i2}, \tilde{\mu}] - \frac{1}{36} m_1 g_1^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right)$$

$$\mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{d}}^{i2}] - \frac{1}{6} g_1^2 y_d^{i1i2} \mathrm{LF}_{1,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}] -$$

$$\frac{1}{6} C_{\phi2^2} g_1^2 y_d^{i1i2} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{d}}^{i2}] + \frac{1}{6} m_1 C_{\phi1\phi2}\tilde{\mu}\, g_1^2 y_d^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{d}}^{i2}] -$$

$$\frac{1}{12} C_{\phi2^2} g_1^2 y_d^{i1i2} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}] + \frac{1}{12} m_1 C_{\phi1\phi2}\tilde{\mu}\, g_1^2 y_d^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}] -$$

$$\frac{3}{2} g_2^2 y_d^{i1i2} \mathrm{LF}_{1,1,1,-1}[m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{3}{4} C_{\phi2^2} g_2^2 y_d^{i1i2} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}] +$$

$$\frac{3}{4} m_2 C_{\phi1\phi2}\tilde{\mu}\, g_2^2 y_d^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}] - \frac{8}{3} m_3 g_3^2 a_d^{i1i2} \mathrm{LF}_{1,1,1,0}[m_3, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}] +$$

$$\frac{2}{3} m_3 g_3^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}] +$$

$$\frac{2}{3} m_3 g_3^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{d}}^{i2}] +$$

$$\frac{1}{18} m_1 g_1^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{i2}, m_1, m_{\tilde{q}}^{i1}] -$$

$$\frac{1}{18} m_1 g_1^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{i2}, m_1, m_{\tilde{q}}^{i1}] -$$

$$\frac{4}{3} m_3 g_3^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{i2}, m_3, m_{\tilde{q}}^{i1}] +$$

$$\frac{4}{3} m_3 g_3^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{i2}, m_3, m_{\tilde{q}}^{i1}] -$$

$$\tilde{\mu}^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] - \frac{1}{2}\tilde{\mu}\, y_d^{pi2} y_u^{i1r}\left(C_{\phi1\phi2}\overline{a_u}^{pr} + C_{\phi2^2}\tilde{\mu}\,\overline{y_u}^{pr}\right)$$

$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] + \frac{1}{2}\tilde{\mu}\, y_d^{pi2} y_u^{i1r}\left(C_{\phi1\phi2}\overline{a_u}^{pr} + C_{\phi2^2}\tilde{\mu}\,\overline{y_u}^{pr}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] +$$

$$\frac{1}{4}\tilde{\mu}\, y_d^{pi2} y_u^{i1r}\left(C_{\phi1\phi2}\overline{a_u}^{pr} + C_{\phi2^2}\tilde{\mu}\,\overline{y_u}^{pr}\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, \tilde{\mu}, m_{\tilde{u}}^{r}] +$$

$$\frac{1}{18} m_1 g_1^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{i2}] -$$

$$\frac{1}{18} m_1 g_1^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{i2}] -$$

$$\frac{4}{3} m_3 g_3^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{d}}^{i2}] +$$

$$\frac{4}{3} m_3 g_3^2\left(C_{\phi2^2} a_d^{i1i2} + C_{\phi1\phi2}\tilde{\mu}\, y_d^{i1i2}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{d}}^{i2}] -$$

$$\frac{1}{2}\tilde{\mu}\, y_d^{pi2} y_u^{i1r}\left(C_{\phi1\phi2}\overline{a_u}^{pr} + C_{\phi2^2}\tilde{\mu}\,\overline{y_u}^{pr}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$$

$$\frac{1}{2}\tilde{\mu}\, y_d^{pi2} y_u^{i1r}\left(C_{\phi1\phi2}\overline{a_u}^{pr} + C_{\phi2^2}\tilde{\mu}\,\overline{y_u}^{pr}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$$

$$\left.\frac{1}{4}\tilde{\mu}\, y_d^{pi2} y_u^{i1r}\left(C_{\phi1\phi2}\overline{a_u}^{pr} + C_{\phi2^2}\tilde{\mu}\,\overline{y_u}^{pr}\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{u}}^{r}, \tilde{\mu}, m_{\tilde{q}}^{p}]\right)$$